A Taxonomy of User Actions on Social Networking Sites

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ABSTRACT

The spread of information within and across Social Networking Sites (SNSs) is increasingly impactful on contemporary society. As information (and misinformation) moves across multiple online platforms, it is important to be able to put these platforms in conversation with one another in order to better understand complex phenomena. This article proposes a taxonomy of actions that are consistent across SNSs to provide researchers and other stakeholders with consistent terminology that enables classifying and comparing user activities over a variety of social media platforms. The proposed taxonomy of actions indicates that although SNSs differentiate themselves in the market and at the level of user experience through unique capabilities and forms of interaction, they can be productively understood as varying means to perform the same set of underlying actions: create, vote, follow, and post.

CCS CONCEPTS

• Collaborative and social computing • Social networking sites • Open source software

KEYWORDS

taxonomy, social networking sites, user behavior

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1 Introduction

User behavior moves information both within and across SNS platforms, but platforms are difficult to compare, as user activities, content, and even basic user properties might not map neatly across them, which creates difficulties in understanding practices that take place across platforms. This is amplified by the artificial divide that leads us to treat current SNSs (those recognizable as social media platforms, including Twitter and Facebook) and content-driven platforms (exemplified historically by LiveJournal and Geocities, but still central to platforms like GitHub) differently, despite their connected ecosystems and shared affordances. Being able to compare diverse platforms using consistent language facilitates understanding a wide variety of phenomena of the contemporary internet where groups coordinate and execute exploits across platforms.

2. Developing the Taxonomy

In developing the taxonomy, we examined several studies geared toward proposing broad categorizations of the activities that take place on social media. For instance, Kietzmann et al. theorized that SNSs have a series of seven functions across platforms: identity, conversation, sharing, presence, relationships, reputation, and groups.[1] For their part, Richthammer et al. have developed categorizations for user data types collected through engagement with various SNSs.[2] Kietzmann et al.'s identity function is the representation of a user's identity, which can be tied to a user's profile.[1] This is the function that corresponds to one of Richthammer et al.'s user-related data types, mandatory data, which is the data users must provide when opening an account on an SNS.[2] This could include a user's name, age, or email address.

Richthammer et al. argue that users can then choose to provide a second category of information, optionally-provided data.[2] These forms of data correspond to Kietzmann et al.'s other functions. Communicating with others via social media falls into the conversation function; this communication might take the form of status updates or direct messages to other users,[1] corresponding to Richthammer et al.'s two types of communication data, private and 1:n communication, which are distinguished from one another based on the audience of a user's post.[2] Users can make connections to other users due to the networking capabilities of social media platforms; the relationships function refers to the ability to relate users to one another via this network.[1] Certain markers like the number of "friends" a user has or the number of "likes" on a post can function to establish a user's reputation on a given platform. Finally, the groups function represents the ability of users to form communities and subcommunities on a platform.[1] Together, these functions offer a comprehensive understanding of the manner in which social media platforms operate, regardless of differences in low-level affordances.

Putting Kietzmann et al. and Richthammer et al in conversation with one another traces out a gap where the conjunction of high level affordances, their functions, and the data provided points to fundamental categories underlying user actions on social media.[1, 2] We theorized four broad user actions that encompass the user activities possible across all SNSs. These actions: *create*, *vote*, *follow*, and *post*, are listed in Figure 1 along with the functions for which users apply each action and the kinds of data a user may be required to provide. Each action operates on a continuum. For example, create encompasses both the creation and deletion of accounts and content spaces. Similarly, post includes both the act of posting content on a social media platform as well as the act of deleting one's content.

Actions	Functions	Data
Create	Identity; presence	Mandatory; Extended profile data
Vote	Reputation	Ratings/interests
Follow	Relationships; Groups	Network
Post	Conversations; presence; sharing	Contextual; private communication data; 1:n communication

Figure 1: Taxonomy of User Actions

Figure 2 illustrates the possibilities for each broad user action as additional context to the other user actions. Developing a visual representation of taxonomical actions can provide additional understanding regarding the sequencing of actions that must occur; *creating* an account necessarily precedes the remainder of the actions in the taxonomy and has therefore been positioned above other actions. In addition, a user cannot *vote* until a *post* exists, so *voting* is considered a tertiary action and is located below the actions *follow* and *post*. Further, Figure 1 lays out the actions in terms of their similarities to one another. For instance, *create* and *post* produce nodes in the network of a given SNS and *vote* is an action on nodes, while *follow* produces edges in the network as it necessarily produces a connection between two users, a user and a content space, or a user and a specific post. Accordingly, *create*, *post*, and *vote* have been colored in green, while *follow* is colored in blue.

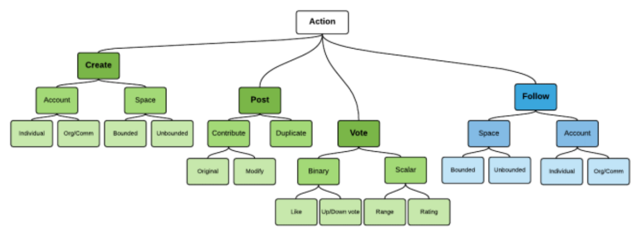

Figure 2: Hierarchical Taxonomy of User Actions

3. Conclusion

While much research has investigated user activities on SNSs, it is difficult to compare user activity across platforms as each platform has its own unique set of affordances and terminology. However, by conceptualizing possible activities on SNSs as a taxonomy, researchers can begin to develop connections between basic user activities regardless of platform. Thinking of user activities as a combination of *create*, *vote*, *follow*, and *post* actions enables the observation of similar patterns of activities across platforms.

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